

Gravity as an Emergent Information Flow in 4D Spacetime

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March 21, 2025

Abstract

This paper derives a new fundamental equation of gravity from first principles, assuming no prior definitions. We propose that gravity is an emergent effect of information flow rather than a fundamental force or curvature. By linking information density gradients to entropy balancing, we establish a novel gravitational equation that does not require mass.

1 Introduction

Gravity has historically been described in two dominant ways:

- **Newtonian gravity** defines it as a force acting at a distance.
- **Einstein's general relativity** describes it as the curvature of spacetime.

However, neither approach explains *why* mass influences spacetime. In this paper, we propose an alternative explanation: gravity is a result of **information flow and entropy gradients** in 4D spacetime.

2 Step 1: Defining Information Flow

We define an **information density function** $I(x^\mu)$ in four-dimensional spacetime:

$$I(x^\mu) = \text{amount of information at } x^\mu \quad (1)$$

where $x^\mu = (ct, x, y, z)$ represents spacetime coordinates.

2.1 Information Gradient

Information does not remain static; it moves in gradients:

$$\nabla I = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}, \frac{\partial I}{\partial z} \right). \quad (2)$$

An object naturally follows the path of steepest information flow, much like water flows down a hill.

3 Step 2: Defining Gravity from Information Flow

We postulate that gravity is the system's response to entropy imbalances:

$$\mathcal{G} = -\nabla I \quad (3)$$

This suggests that **gravity is not a force, but an emergent phenomenon of entropy redistribution.**

4 Step 3: Including Spacetime Effects

Since information propagates through both space and time, we extend our model using the **d'Alembertian operator**:

$$\square I = \frac{c^4}{8\pi G} R \quad (4)$$

where:

- $\square I = \nabla^\mu \nabla_\mu I$ is the wave operator applied to information density.
- R is the Ricci scalar curvature, linking information to spacetime geometry.

5 Step 4: Linking Gravity to Entropy

Since entropy S is related to information density, we use the standard entropy formula:

$$S = k_B \log W \quad (5)$$

where W represents the number of possible states of a system.

Taking the second derivative of entropy over spacetime gives:

$$\mathcal{G} = \frac{c^4}{8\pi G} \nabla^2 S \quad (6)$$

This equation means:

- *Mass is not required*—any high-entropy zone generates gravity-like effects.
- *Gravity emerges purely from information redistribution, not from mass-energy.*

6 Final Equation: The True Law of Gravity

From the previous derivations, the final equation governing gravity as information flow is:

$$\mathcal{G} = -\nabla I = \frac{c^4}{8\pi G} \nabla^2 S. \quad (7)$$

7 Implications and Testable Predictions

This equation leads to several testable predictions:

- Gravity should exist in **high-entropy vacuums** even when no mass is present.
- **Time dilation should depend on entropy flow**, not just mass.
- **Black holes should behave as entropy processors**, not singularities.
- **Dark matter effects could be explained** as delayed entropy corrections.

8 Conclusion

This paper derives gravity from scratch using information theory. The final equation suggests that:

- Gravity is an **entropy-driven effect**, not a fundamental force.
- Mass is not required for gravity to exist.
- Spacetime is shaped by information flow rather than by mass-energy alone.

References

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